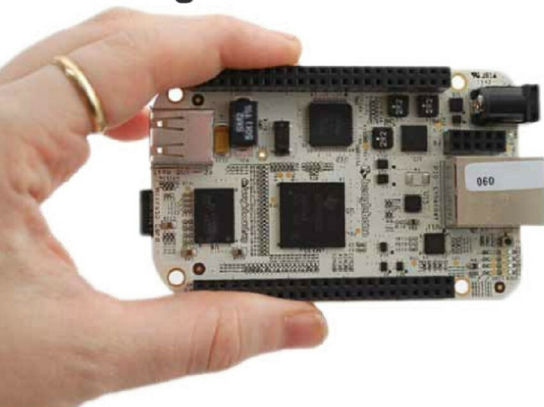


Getting Started with the Beaglebone



Over the next couple of issues, this series aims to make a novice well-versed with the Beaglebone. It will cover various topics such as porting different OSs to the Beaglebone, a few tinkering projects, etc. This first article gives a brief overview of the Beaglebone and how to toggle a user LED on it.

The Beaglebone is a low-cost credit-card-sized Linux computer that runs software such as Android and Ubuntu. With plenty of I/O and processing power for real-time analysis provided by the TI Sitara AM335x ARM Cortex-A8 processor, Beaglebone can be complemented with a cape, which is a plug-in board (similar to the Arduino ‘shield’), to augment functionality. Here are some of its cool features:

- 720 MHz super-scalar ARM Cortex-A8 (ARMv7a) processor
- 3D graphics accelerator
- ARM Cortex-M3 for power management
- 2x Programmable Realtime Unit 32-bit RISC GPUs
- USB client: power, debug and device
- USB host
- Ethernet port
- 2x 46 pin headers
- 2x I2C, 5x UART, I2S, SPI, CAN, 66x 3.3V GPIO and 7x ADC
- 4 GB microSD card with Angstrom distribution
- Cloud9 IDE on Node.js with the Bonescript library
- Size: 8.6 cm x 5.3 cm (3.4" x 2.1")
- Single-cable development environment with built-in FTDI-based serial/JTAG and an on-board hub to give the same cable simultaneous access to a USB device port on the target processor.
- Industry standard 3.3V I/Os on the expansion headers, with easy-to-use 0.254 cm (0.1") spacing.

Purchasing hardware and capes

Regarding buying a Beaglebone, there are a lot of distributors;

pick the one nearest to you from <http://beagleboard.org/buy>. The cost of the Beaglebone is currently \$89, for which you get (in the box) the Beaglebone, a USB cable, and a 4 GB micro SD card.

Currently, there are over 20 capes for DVI-D, VGA, LCD, motor control, prototyping, the battery and more. These can be purchased from <http://BeagleBoneCapes.com/>.

OS support

Though the Beaglebone ships with a 4 GB SD card with a working Angstrom image, it supports a wide variety of OS' such as Ubuntu, Debian, Android, Angstrom, Fedora, Arch Linux, Sabayon, etc, that you can use as alternatives. You can also download the latest Angstrom image for the Beaglebone from <http://downloads.angstrom-distribution.org/demo/beaglebone/>. At the time of writing, the latest image available was ‘Angstrom-Cloud9-IDE-GNOME-eglibc-ipk-v2012.05-beaglebone-2012.11.22.img.xz’. To update the image on the SD card, insert your SD card into the computer, but make sure that it is not mounted. Now, identify the correct raw device name (like `/dev/sde`—not the partition, such as `/dev/sde1`). You can do this by opening the Ubuntu disk utility and clicking the SD card. Now, in the terminal, navigate to the folder where you downloaded the image and run the following command to write the image to the SD card:

```
xz -dkc Angstrom-Cloud9-IDE-GNOME-eglibc-ipk-v2012.05-
beaglebone-2012.11.22.img.xz >/dev/sdX
sync
```